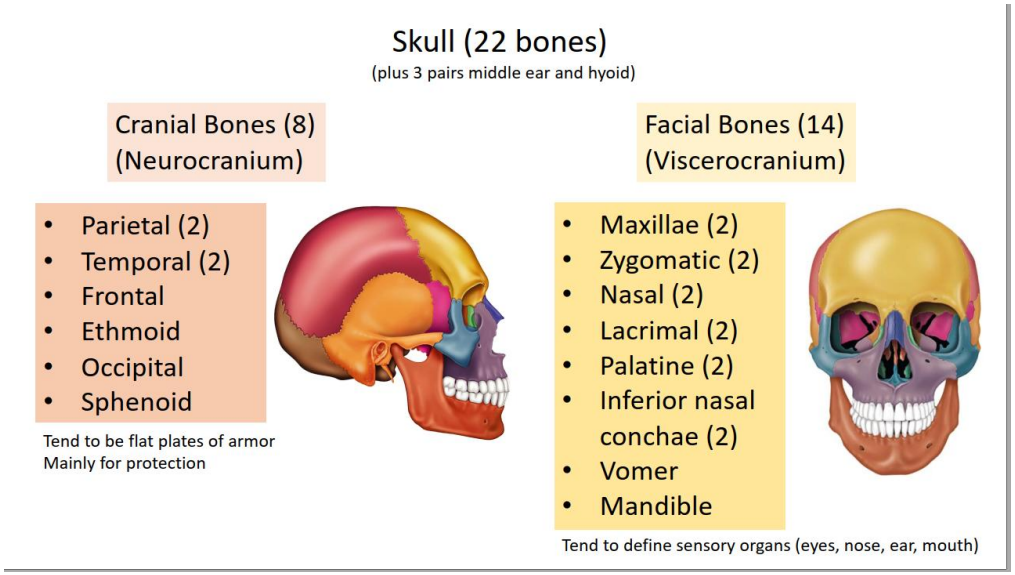
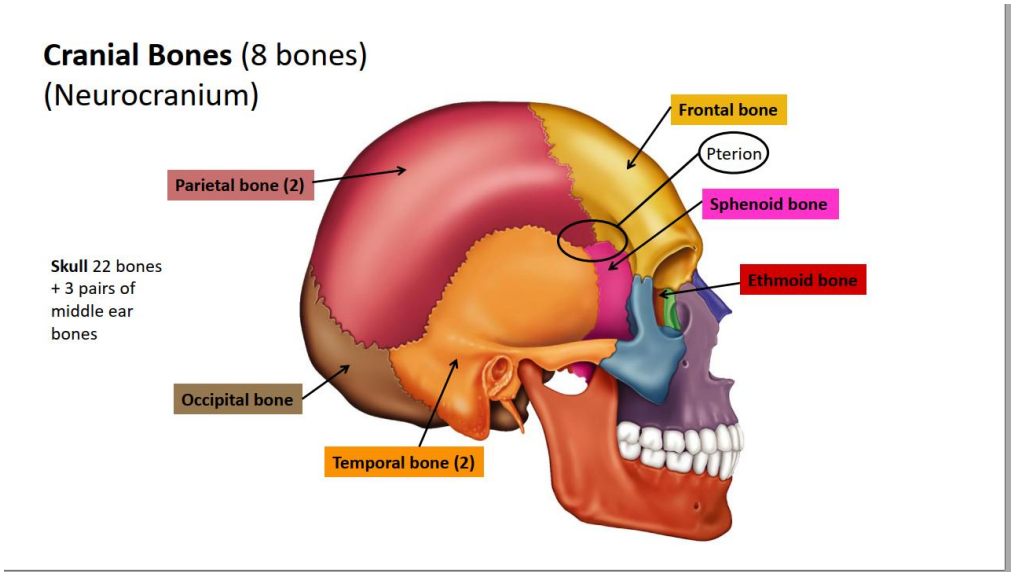
Week #12: [Cranial Cavity]

Osteology
Meninges
Dura
Blood Supply
Cranial Hematoma

Cranial Cavity I: Osteology





I. The Skull or Cranium – Can be divided into 2 parts: the viscerocranium & neurocranium

a. Orbitomeatal plane – horizontal plane extending from the inferior margin of the orbit & the superior margin of the external acoustic opening of the external acoustic meatus (anat. position)
b. Most calvarial bones are united by fibrous interlocking sutures.
c. Several bones of the cranium are pneumatized bones containing air spaces (air sacs or large sinuses), presumably to decrease their weight.

II. Viscerocranium – skeleton of the face (covered previously in year 1)

a. Bones: Mandible (1), ethmoid (1), vomer (1), zygomatic (2), maxillae (2), palatine (2), lacrimal (2), nasal (2), & inferior nasal conchae (2)

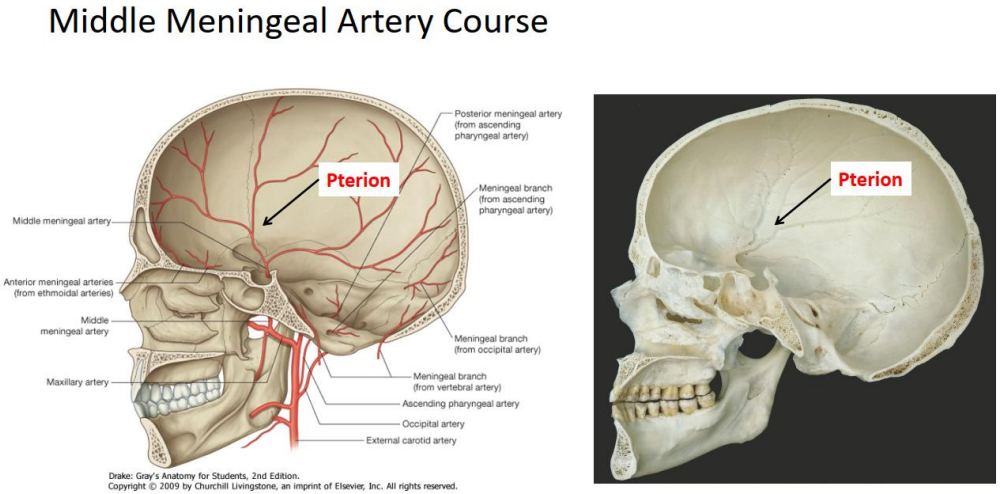
III. Neurocranium – cranial vault or brain case

a. Bones: Frontal (1), occipital (1), sphenoid (1), ethmoid (1), parietal (2), temporal (2)
    i. Cranial sutures: non-mobile articulations between skull bones
        1. ***Coronal suture*** – between the frontal & parietal (2) bones
        2. ***Sagittal suture*** – in the midline, between the two parietal bones
        3. ***Lambdoid suture*** (looks like a "square") – between the parietal (2) & occipital bones
        4. ***Squamosal suture*** – between temporal & parietal bones
        5. ***Frontal suture*** – originally divided the frontal bones of the fetus; a persistent frontal suture or remnant of it is called a metopic suture in adults.

6. ***Bregma*** - intersection of sagittal & coronal sutures
7. ***Lambda*** – intersection of lambdoid & sagittal sutures
8. ***Pterion*** - intersection of frontal, parietal, sphenoid & temporal bones("H-shaped")

B. External Features:

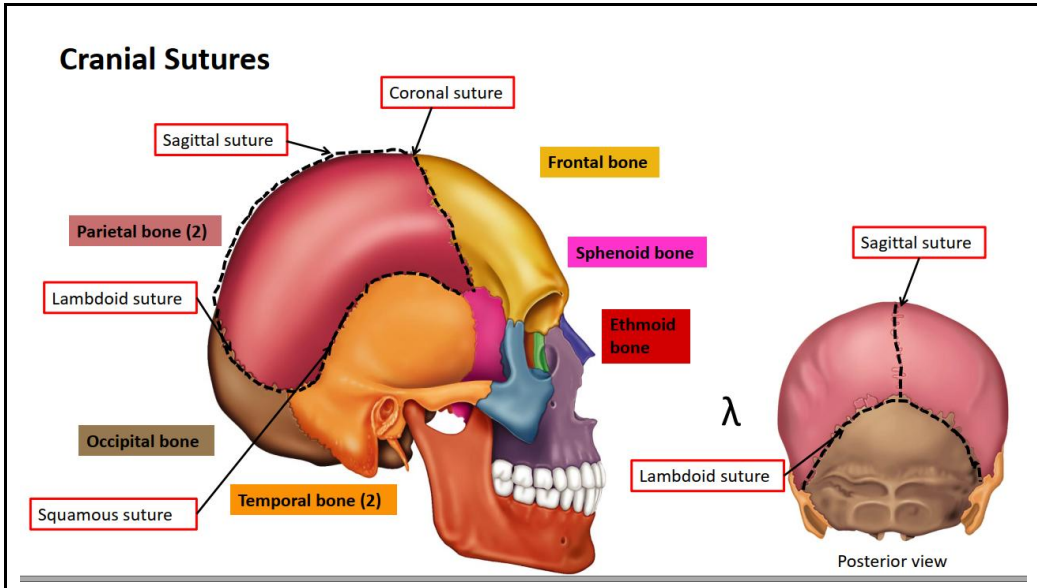
a. On the lateral side of the skull, the temporal fossa is a shallow depression bounded by superior & inferior temporal lines and inferiorly by the zygomatic arch. The infratemporal fossa is an irregular space inferior & deep to the zygomatic arch & mandible and posterior to the maxilla.
b. ***External acoustic opening*** – entrance to ***external acoustic meatus***, which leads to the tympanic membrane
c. ***External occipital protuberance*** – easily palpable elevation in the median plane on the posterior aspect of the head. The superior nuchal line extends laterally from each side, marking the superior limit of the neck; the inferior nuchal line is less distinct.
d. **Mastoid process** – conical prominence off temporal bone; lateral to the styloid process.
e. **Styloid process** – sharp spine projecting from inferior surface of temporal bone; anterior to stylomastoid foramen, which transmits CN VII & stylomastoid artery
f. **Occipital condyles** – where cranium articulates with vertebral column

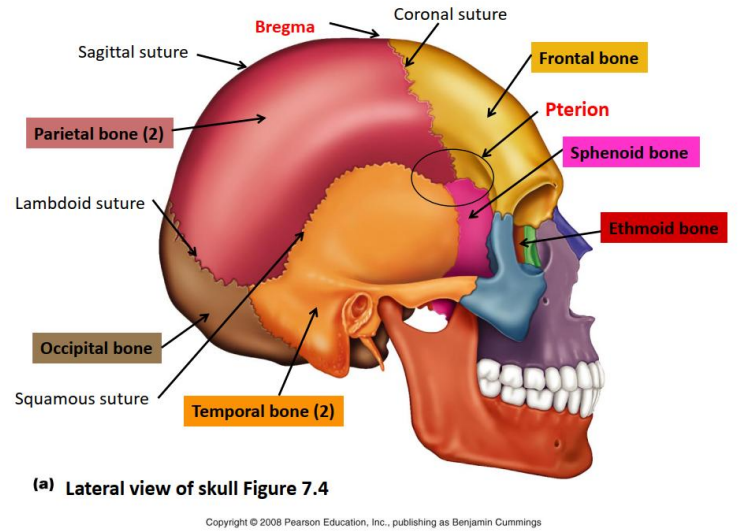


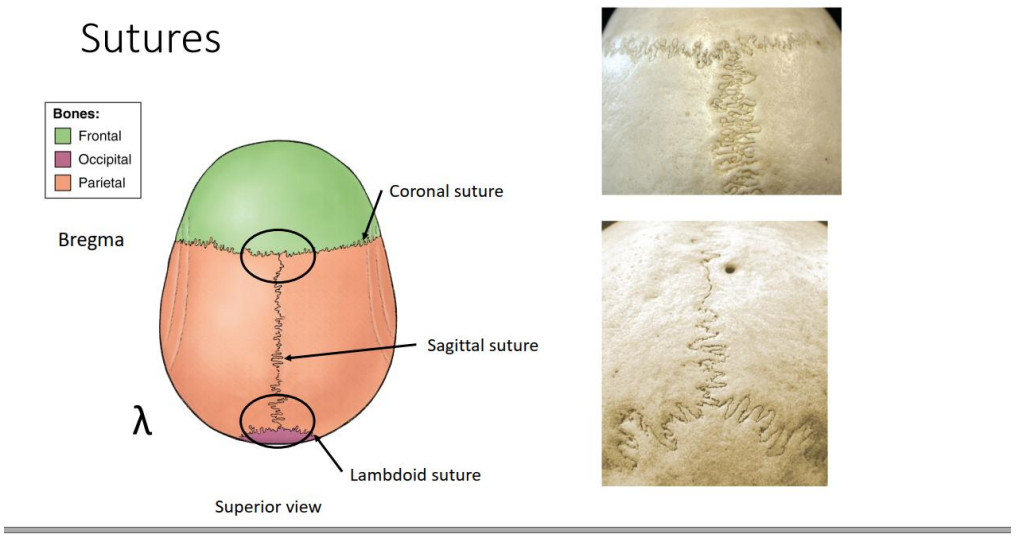
C. Calvaria - "Skull cap" or top of the neurocranium

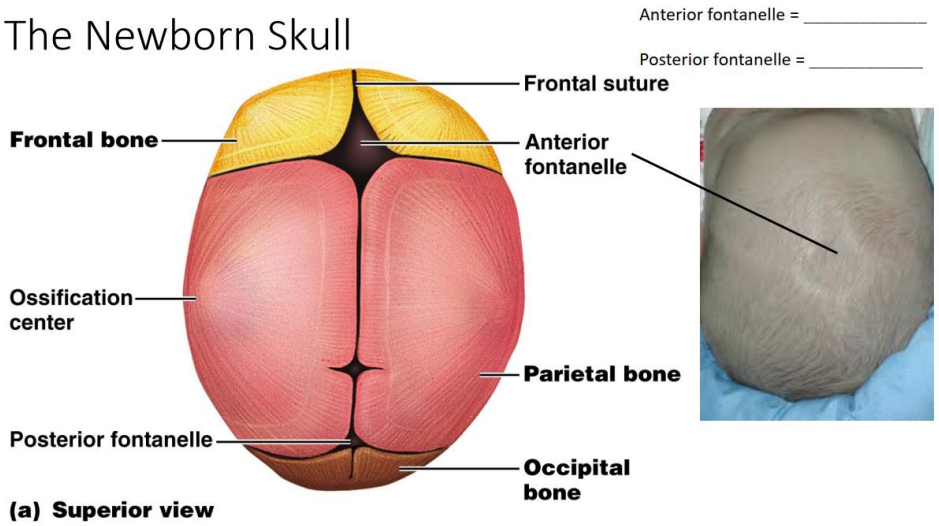
a. The superior surface of the calvaria is composed of the frontal bone (anteriorly), 2 parietal bones (laterally), occipital bone (posteriorly). NOTE: Sutural bones (accessory bones) may also be present.
b. Inferior surface of the calvaria:
   i. **Sagittal groove** - attachment of the superior sagittal sinus; widens posteriorly
   ii. **Vascular grooves** – of the ***middle meningeal artery*** (in parietal bones)
   iii. **Granular foveolae** - pits or depressions in the calvaria formed by *arachnoid granulations*, where cerebrospinal fluid (CSF) is returned to the dural venous sinuses. They occur mostly along the superior sagittal sinus.

Clinical Note: A ***fracture at the pterion*** can be life-threatening because it overlies the anterior branches of the ***middle meningeal vessels***, which are running in the grooves on the internal aspect of the lateral wall of the cranium. A fracture of these thin bones could rupture the middle meningeal artery resulting in an ***epidural hematoma.***









D. Development of the Skull

1. Bones of the calvaria & some parts of the cranial base develop by ***intramembranous ossification***; most parts of the cranial base develop by ***endochondral ossification.***
2. The bones of the skull complete their development & ossification postnatally. Thus, infants have an undeveloped skull & face at birth and early infancy.
3. **Fontanelles** are membranous gaps between bones of the skull in a fetus or infant, they remain connected by soft fibrous sutures at birth permitting skull deformation in the birth canal (molding) and cranial expansion during early childhood. They usually close gradually, often by 8 months to 1 year of age.
    a. These include: **anterior** (corresponds to bregma), **posterior** (corresponds to lambda), and paired **sphenoid & mastoid fontanelles.**
    b. Clinically, palpation of the fontanelles allows evaluation of normal developmental signs such as hydration, intracranial pressure, and bone growth.

Compare these images of radiographs from children. One is normal the other is not. What has happened to cause this problem?



Normal



Elongated

premature closure of sutures- not allowing brain to expand in proper shape

Craniosynostosis



https://www.drruas.com

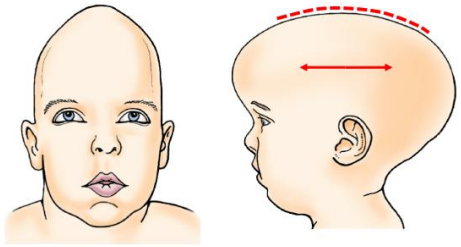


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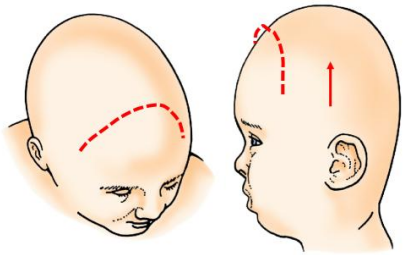


https://cranialtherapycenters.com

Craniosynostosis: Premature closing of sutures



Sagittal Suture

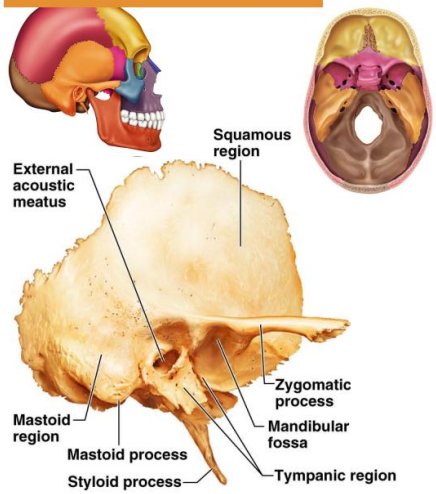


Coronal Suture

Clinical Note: **Craniosynostosis**, or premature closure of the cranial sutures, results in several types of cranial malformations.

- ***Scaphocephaly*** is a premature closure of the sagittal suture. Other sutures, including the coronal and metopic sutures, may be involved. While the first sign is an abnormal head shape, other symptoms may develop including increased intracranial pressure, developmental delays leading to mental retardation, seizures, & blindness if not recognized & surgically corrected.

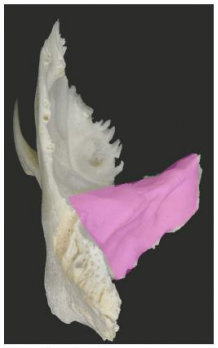
Temporal Bone



4 Regions:

1. Squamous
2. Mastoid
3. Temporal
4. Petrous (inside cc)


Lateral view


Posterior view

E. Three Layers of the Cranial Wall

1. Most flat bones of the skull have **two lamina** (or layers) of compact bone that are separated by the **diploë**, a layer of spongy bone.

❖ **Clinical Note:** The inner layer of compact bone is distinctly thinner and more prone to injury. In fact, the inner layer can fracture with the outer lamina still intact.

2. The diploë contains bone marrow and canals that transmit the **diploic veins**, which drain either into the veins of the scalp or dural venous sinuses.
3. Veins that pass through all three layers of the skull and connect the dural venous sinuses with veins outside the skull are called **emissary veins**. They pass through ***emissary foramina***. These veins are highly variable in number & location.
4. The thickness of the cranium varies among regions, which is best visualized when holding the skull up to the light. The cranium is usually thinnest in regions covered by muscles.

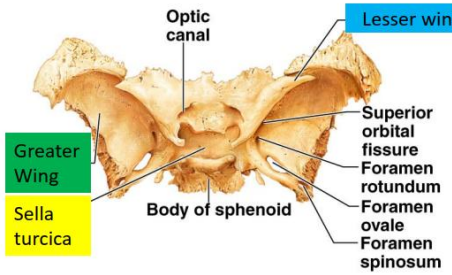
❖ **Clinical Note:** Hard blows to the head at thin areas of the skull are likely to result in **fractures to the calvaria**. *Depressed fractures* result in a fragment of bone being depressed inward, compressing and/or injuring the brain. In *comminuted fractures*, the bone breaks into several pieces. *Linear fractures* (the most frequent) occur at the point of impact, but fracture lines often radiate away from it in two or more directions. A *contrecoup fracture* occurs on the opposite side from the trauma.

Sphenoid Bone





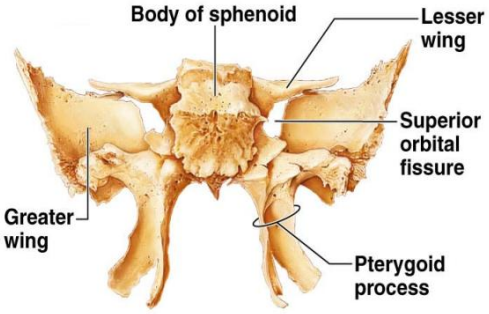


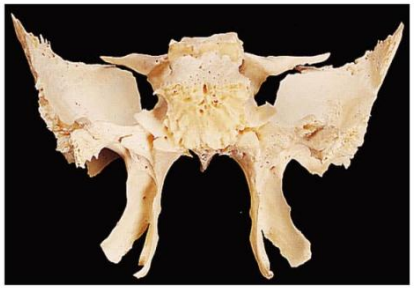




Sphenoid Bone Posterior View



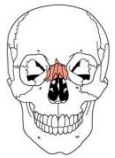


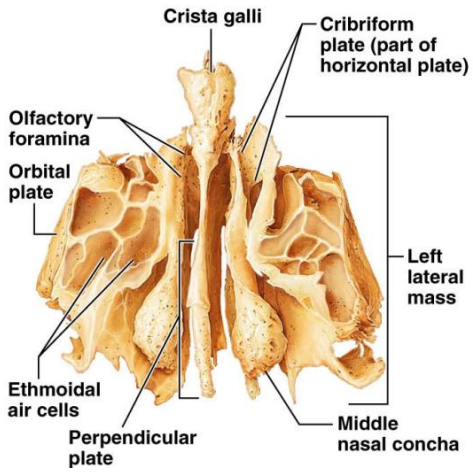


(b) Posterior view

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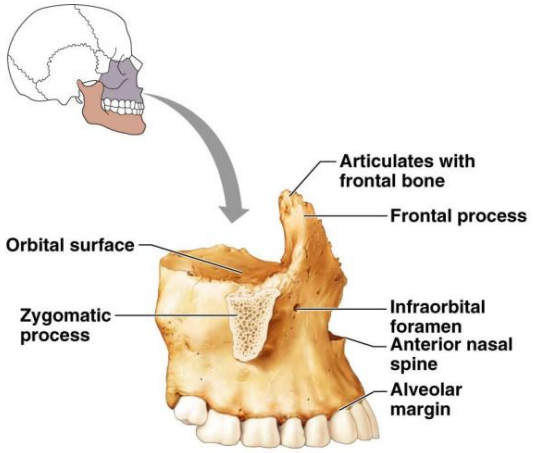
Ethmoid Bone







Maxilla



(b) Maxilla

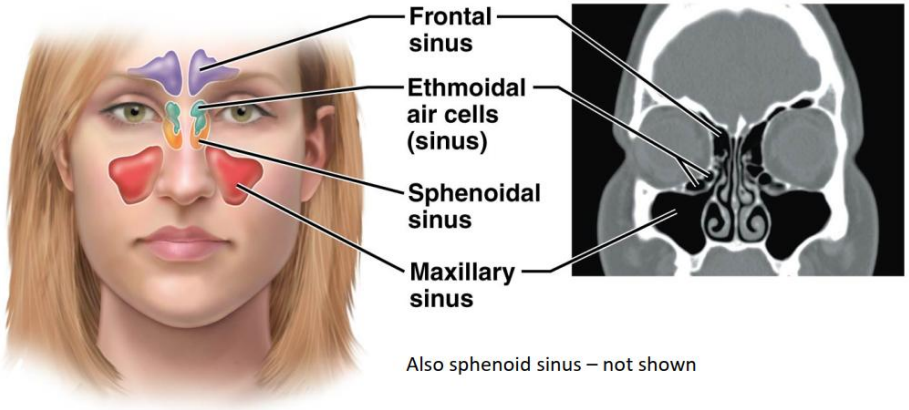
Infraorbital foramen
– V2 maxillary division of CN 5



(c) Maxilla, right lateral view

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Paranasal Sinuses

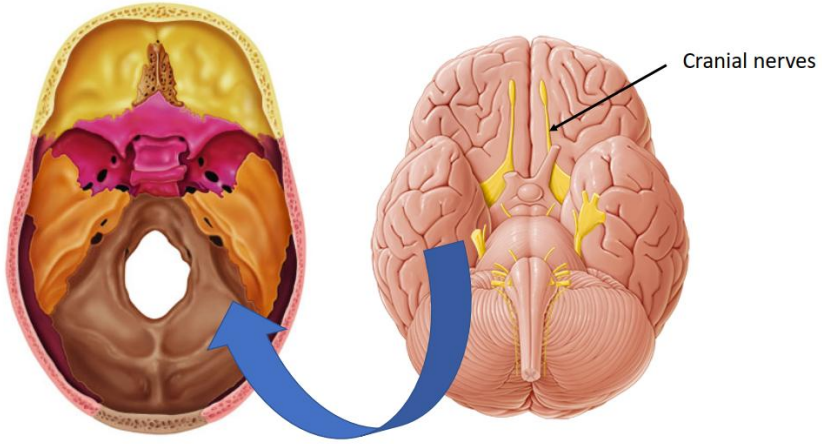


Also sphenoid sinus – not shown

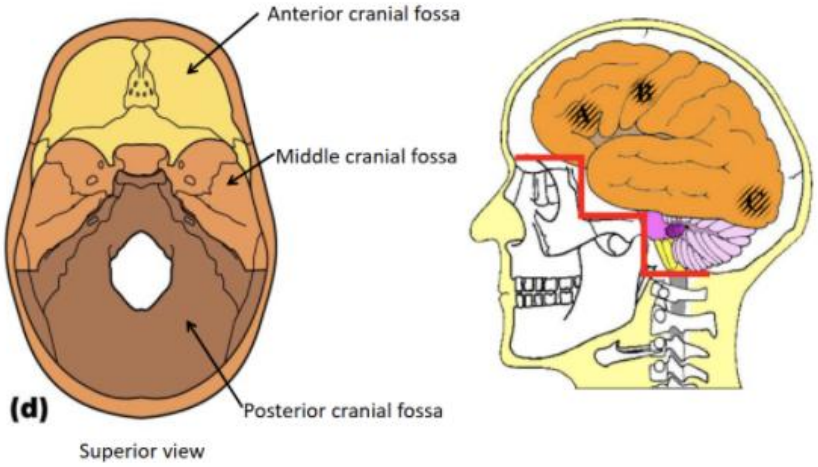
(a) Anterior aspect

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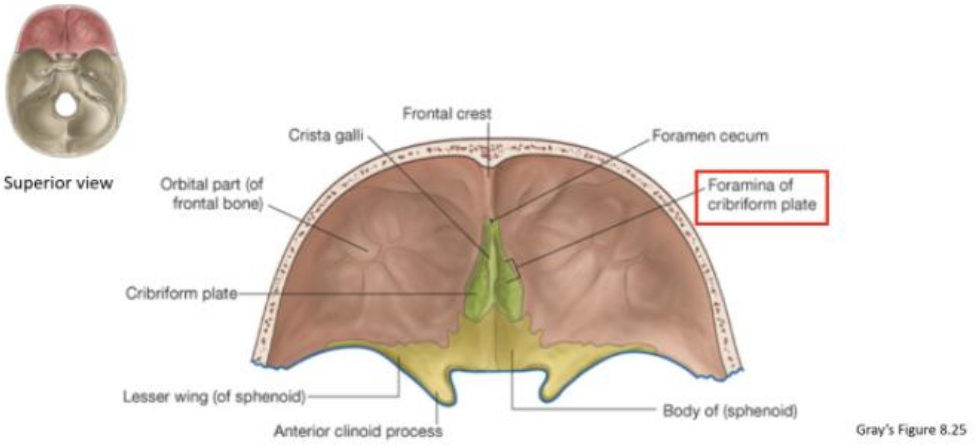
Cranial Cavity



Cranial Fossae

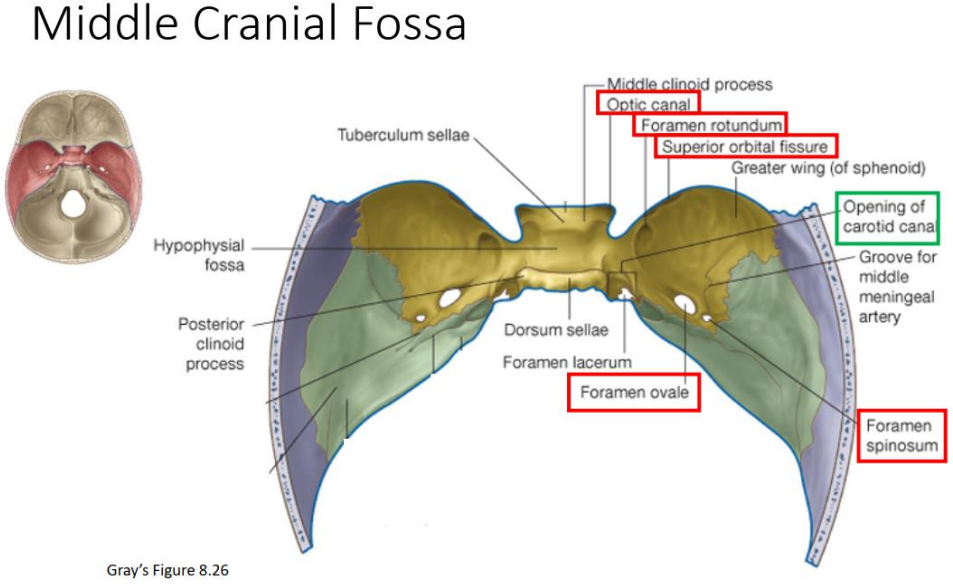


Anterior Cranial Fossa



Cranial Fossae – depressions formed along the floor of the cranial cavity

- **A. Anterior Cranial Fossa:**
  - a. ***Bones:*** Frontal, ethmoidal, and sphenoid
  - b. Houses the frontal cerebral lobes of the brain
  - ***c. Features:***
    - **i. Frontal**
      1. ***Orbital plate*** – forms roof of orbit
      2. ***Frontal crest*** – median bony extension of frontal bone
      3. ***Foramen cecum*** – gave passage to fetal vessels
    - **ii. Ethmoid**
      1. ***Crista galli*** – thick, median ridge of bone
      2. ***Cribriform plate & foramina*** – transmit CN I
    - **iii. Sphenoid**
      1. ***Lesser wings*** – arise from upper & anterior parts of sphenoid body and project laterally, ending in sharp points
      2. ***Anterior clinoid processes*** – bony projections off sphenoidal ridge
  - d. Anterior & middle cranial fossae are separated by the sphenoidal ridge/crest.

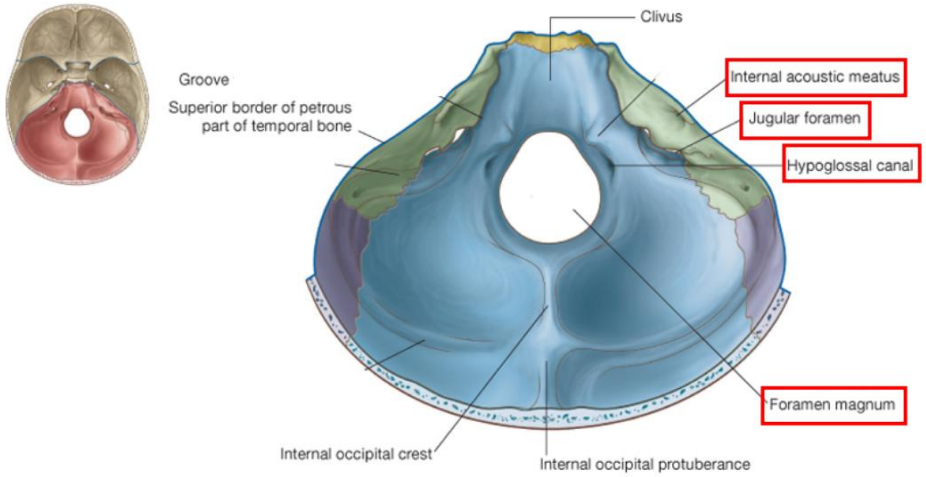


B. Middle Cranial Fossa:

1. Bones: Sphenoid (body & greater wing) and Temporal (squamous & petrous portions)
2. Features of the Central Part: houses pituitary gland
    1. **Sphenoidal limbus** (central anterior boundary of fossa)
    2. **Prechiasmatic sulcus** (groove between optic canals for ***optic chiasm***)
    3. ***Optic canals – transmit CN II & ophthalmic artery***
    4. **Sella turcica**: bony formation on superior surface of body of sphenoid
        1. **Tuberculum sellae** (horn of saddle)
        2. **Dorsum sellae (back of saddle) with posterior clinoid processes**

    iii. **Hypophysial fossa** (pituitary fossa)
    1. **Carotid groove –** groove for internal carotid artery & cavernous sinus
3. Features of the Lateral Part: houses the temporal lobes of brain
    1. **Crest** of the petrous portion of temporal bone
    2. **Tegmen tympani –** roof of tympanic cavity
    3. **Foramen lacerum** (artifact; closed off by cartilage plate in the living)
    4. **Carotid canal** – transmits the internal carotid artery into the cranial cavity
    5. ***Superior orbital fissure – transmits CN III, IV, V 1, & VI***
    6. **Foramen rotundum** – transmits maxillary nerve (***CN V2***)
    7. **Foramen ovale** – transmits mandibular nerve (***CN V3***)
    8. **Foramen spinosum** – transmits ***middle meningeal artery***
4. Middle & posterior cranial fossae are separated by the ***crest of the petrous part of the temporal bone***.

Posterior Cranial Fossa

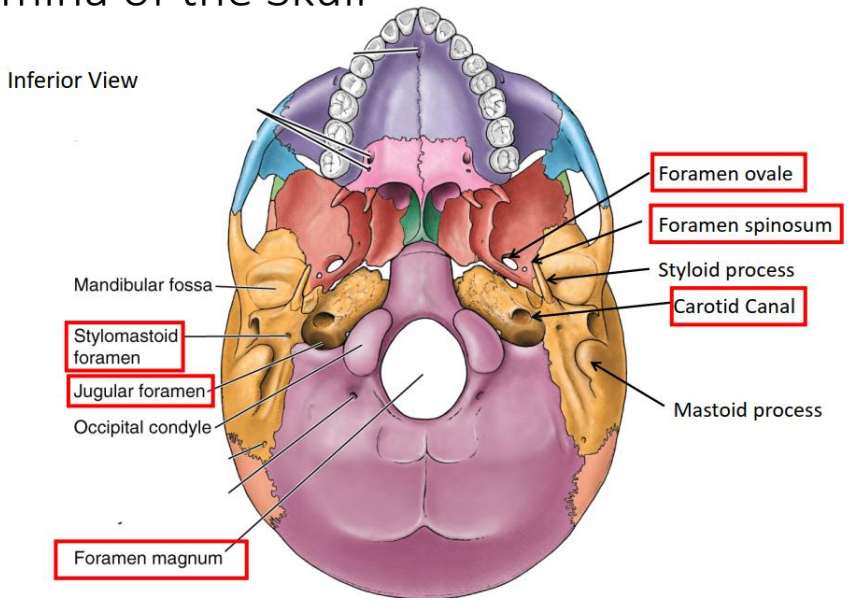


Gray's Figure 8.27

C. Posterior Cranial Fossa

1. Bones: Occipital, sphenoid, temporal, and parietal bones
2. Houses the cerebellum, pons, medulla, & cisterna magna
3. Features:
    1. **Clivus** – a marked incline leading from the foramen magnum
    2. **Internal occipital crest & protuberance**
    3. **Grooves for the: Transverse sinus, Sigmoid sinus, & Inf. & superior petrosal sinuses**
4. Foramina:
    1. **Foramen magnum** – transmits ***CN XI***, vertebral arteries, spinal cord & meninges
    2. ***Hypoglossal canal – transmits CN XII***
    3. **Jugular Foramen** – transmits ***CN IX, X, & XI***; sigmoid sinus → internal jugular vein
    4. ***Internal acoustic meatus – transmits CN VII & VIII***

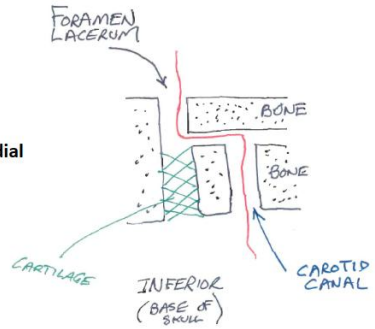
Foramina of the Skull



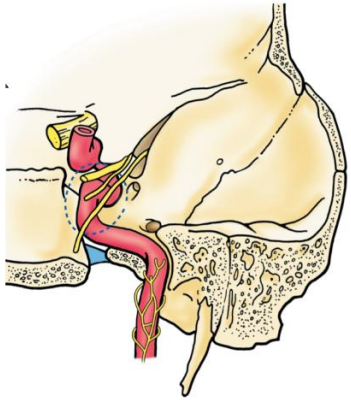
Path of the Internal Carotid Artery (ICA)

a. The Internal Carotid artery (ICA) enters the skull through the ***external opening of the carotid canal*** on the external base of skull, anterior to the jugular foramen.
b. Runs through the ***carotid canal*** within the skull, changing course from vertical to horizontal, as it is directed anteriorly & medially.
c. Exits the skull to enter the middle cranial fossa via the ***internal opening of the carotid canal***, where the ICA passes superiorly over the foramen lacerum.
d. Passes through the ***carotid grooves*** (lateral to sella turcica) as it courses anteriorly through the cavernous sinuses.
e. Makes a 180° turn inferior to the anterior clinoid processes to head posteriorly.
f. Joins the ***cerebral arterial circle*** (ofWillis).

Carotid Canal



Medial

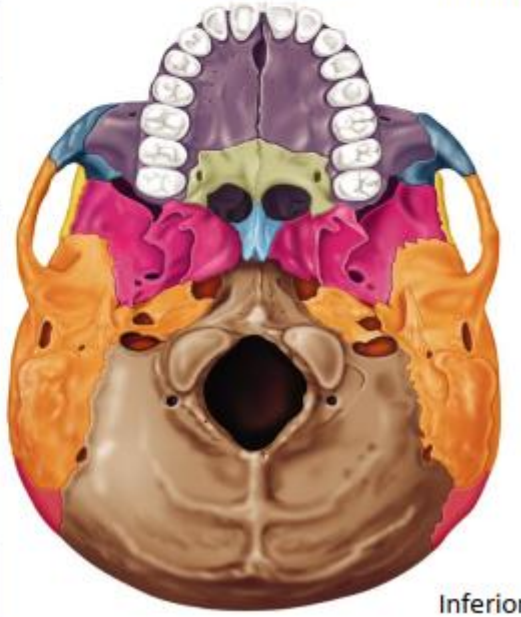


Lateral

Foramen	Structure
Cribriform plate	CN I - Olfactory nerve filaments
Optic foramen (canal)	CN II - Optic nerve and ophthalmic artery
Superior orbital fissure	CN III - Oculomotor nerve CN IV - Trochlear nerve CN V_1 - Trigeminal nerve (ophthalmic division) CN VI - Abducens nerve
Foramen rotundum	CN V_2 - Trigeminal nerve (maxillary division)
Foramen ovale	CN V_3 - Trigeminal nerve (mandibular division)
Foramen spinosum	Middle meningeal artery
Foramen lacerum	Internal carotid artery
Internal auditory meatus	CN VII - Facial Nerve CN VIII - Vestibulocochlear nerve
Jugular foramen	Jugular vein CN IX - Glossopharyngeal nerve CN X - Vagus nerve CN XI - Accessory nerve
Hypoglossal canal	CN XII - Hypoglossal nerve
Foramen magnum	Brain stem and spinal cord, Vertebral arteries



Superior view

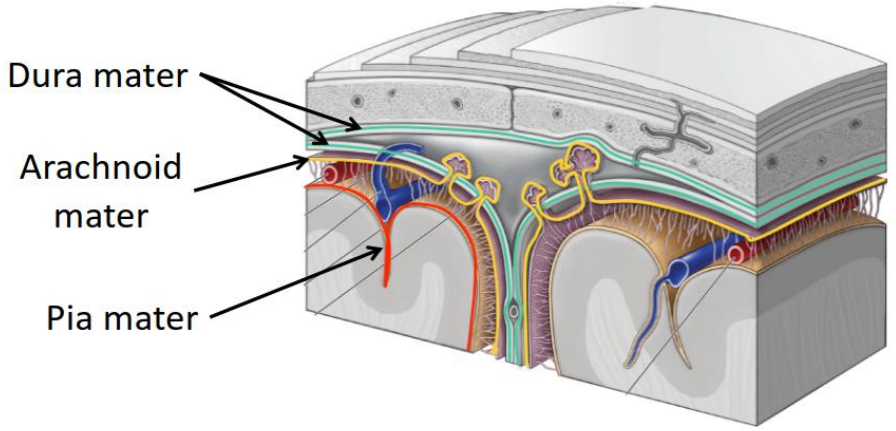


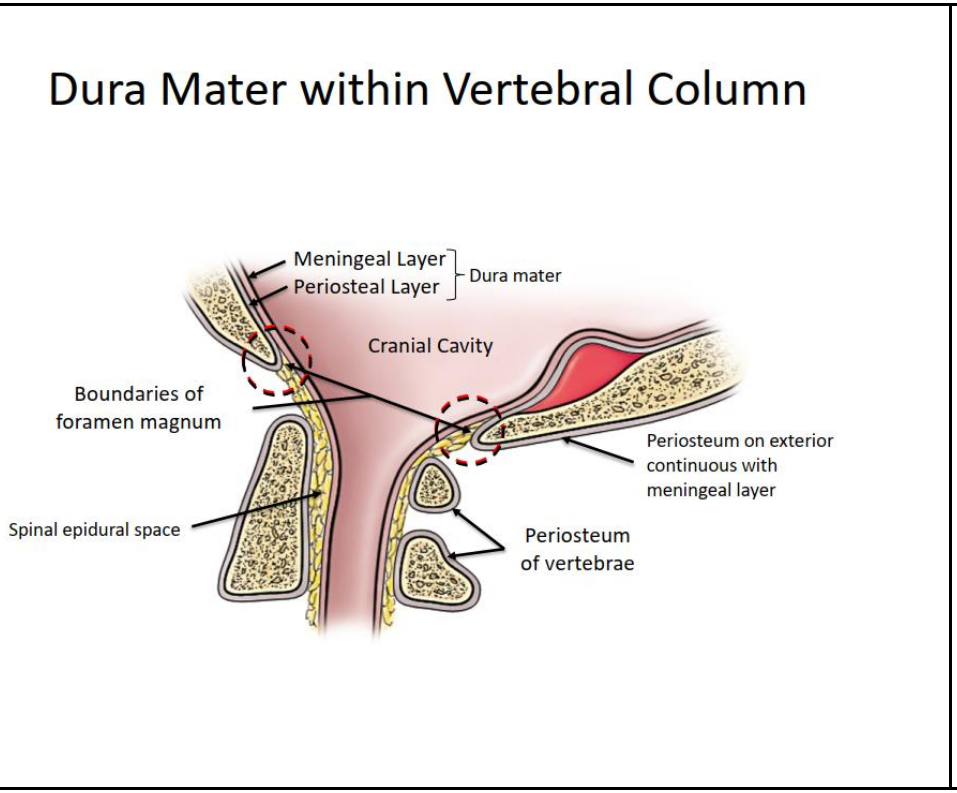
Inferior view

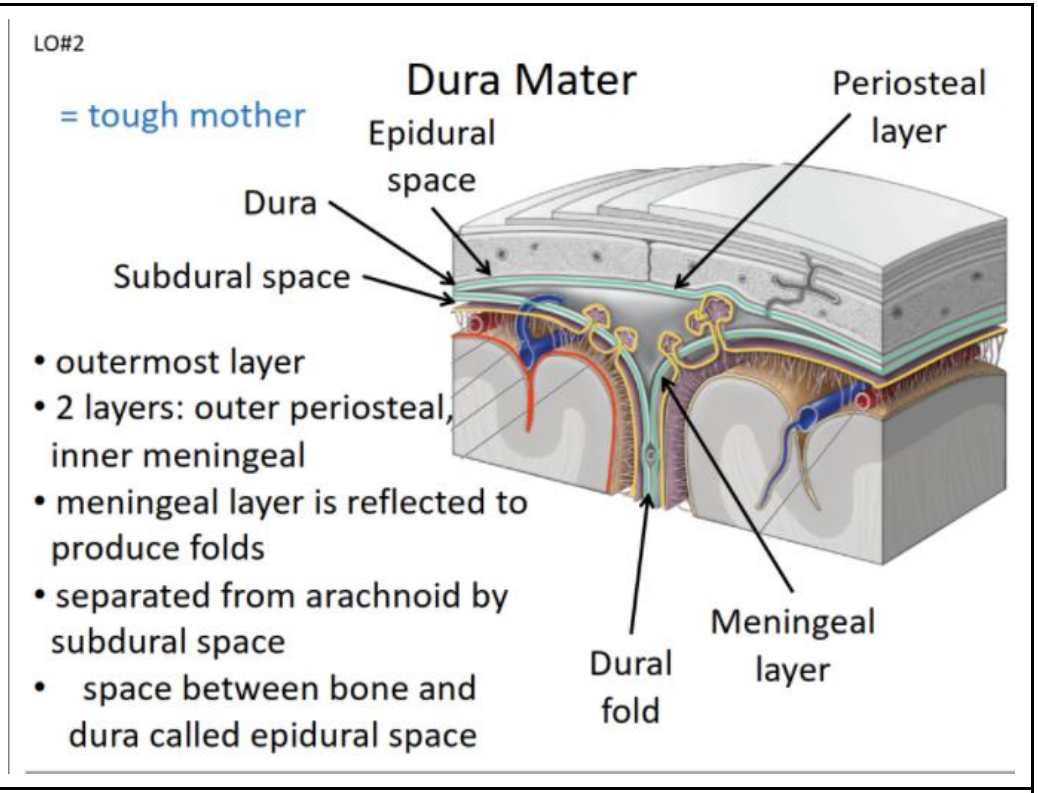
LO#2

Meninges

- Cover the brain
- Continuous with spinal cord
- 3 layers







VI. Cranial Meninges – coverings of the brain immediately internal to the cranium

A. Functions:

1. Protect the brain
2. Serve as supporting framework for arteries, veins, and dural venous sinuses
3. Enclose the subarachnoid space, where blood vessels & CSF reside.

B. Consist of 3 layers:

1. **Dura mater** – tough, thick external fibrous layer
2. **Arachnoid mater** – avascular, web-like, thin intermediate layer
3. **Pia mater** – delicate, internal vascular layer tightly adherent to the brain

C. Dura Mater ("Tough Mother")

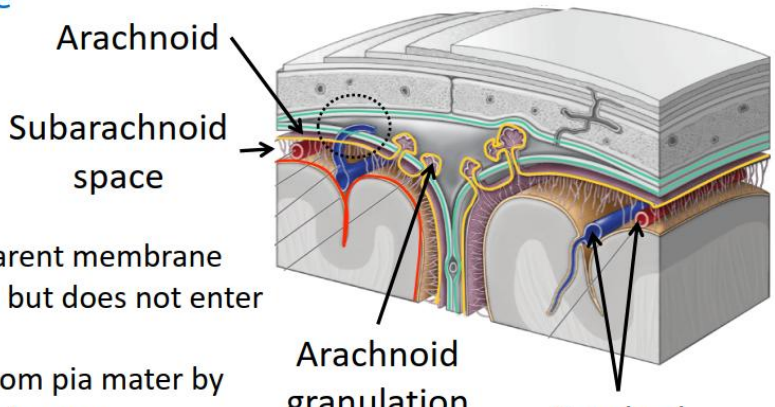
1. Composed of two layers that are (usually) fused.
    a. **Periosteal (endosteal) layer of dura** – external layer; internal periosteum of the skull that is continuous with the external periosteal layer through the various foramina, but NOT continuous with the dura mater of the spinal cord.
    b. **Meningeal layer of dura** – internal layer; continuous with the dura mater of the spinal cord at the foramen magnum, but NOT continuous with the spinal epidural space.

Clinical Note: Epidural anesthesia can NOT ascend to enter the skull.
(More on dura below)

LO#2

Arachnoid Mater

= spider-like

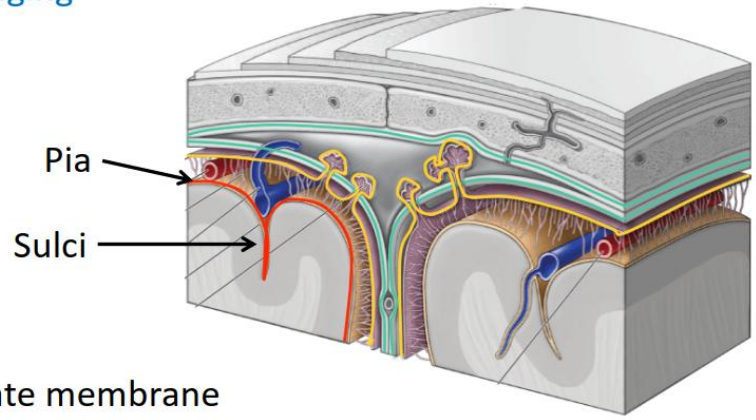


- thin, transparent membrane
- covers brain but does not enter sulci
- separated from pia mater by subarachnoid space
- subarachnoid space contains CSF and cerebral arteries and veins
- forms arachnoid granulations

LO#2

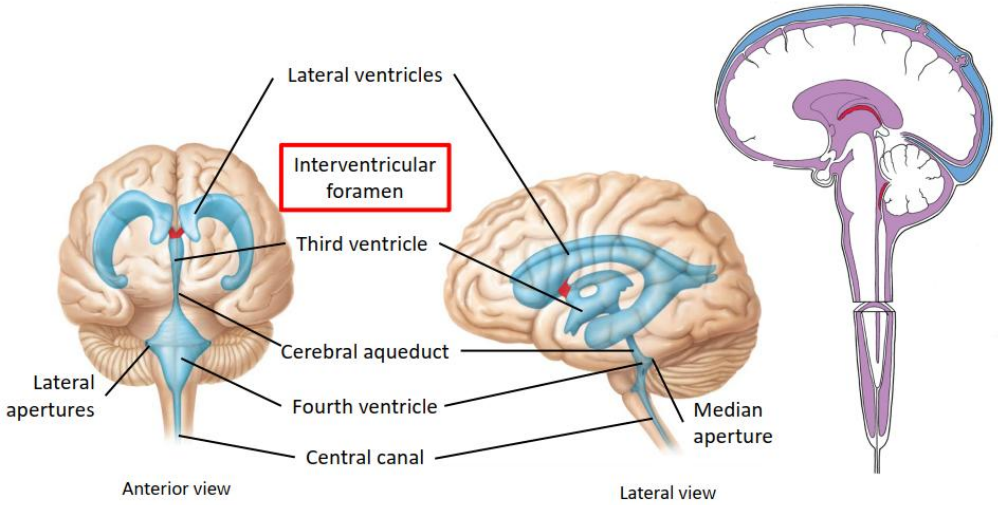
Pia Mater

= faithful, clinging



- thin, delicate membrane
- adherent to surface of brain and enters sulci

Ventricles



- CSF formed by the choroid plexus in the third and fourth ventricles
- Circulates through the ventricles and into the subarachnoid space via the median and lateral apertures
- Drains into venous circulation at superior sagittal sinus through arachnoid granulations

LO#2

Meningitis =

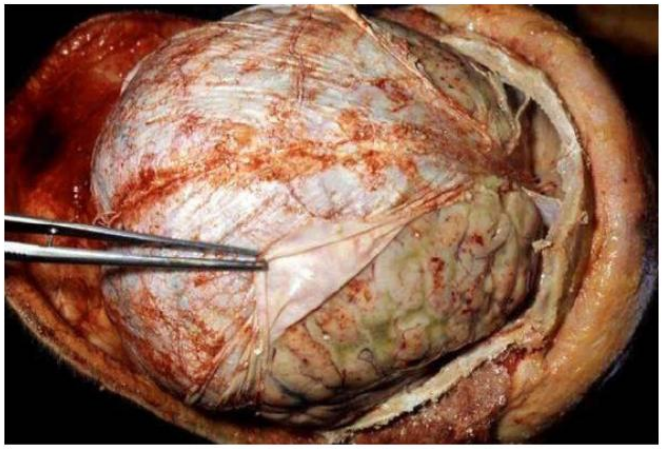
Inflammation of the meninges

Symptoms:

- Headache
- Fever
- Drowsiness

Can lead to:

- Infection and sepsis
- Coma
- Death

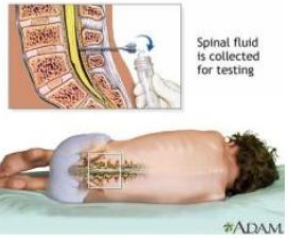


LO#2

Detecting Meningitis



Kernig's Sign



Lumbar Puncture

Kernig's Sign

- Straight leg raises causes marked pain/discomfort

Brudzinski's Sign

- Involuntary lifting of legs when lifting patient's head

Nuchal Rigidity

- Inability to flex neck due to rigid muscles

Lumbar Puncture

- Take sample of CSF
- From where?

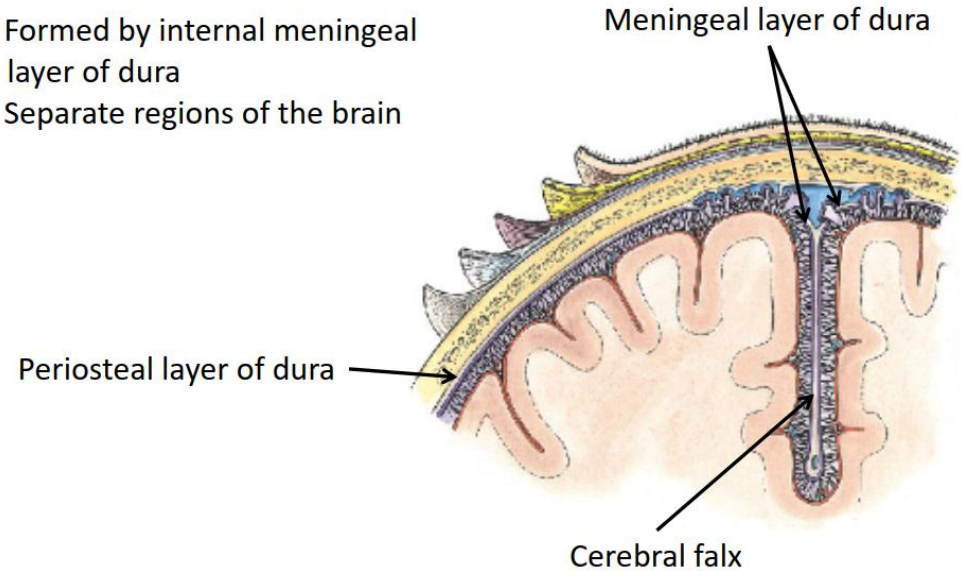
Main Points - Meninges

- Meninges cover and help protect the brain
- 3 layers of meninges:
  - Outermost layer is dura mater
  - Middle layer is arachnoid mater
  - Innermost layer is pia mater
- Potential space between dura mater and skull is epidural space
- Potential space between dura mater and arachnoid mater is subdural space
- Real space between arachnoid mater and pia mater is subarachnoid space – where cerebral arteries and veins and CSF run
- Meningitis is inflammation of meninges

LO#2

Dural Reflections

- Formed by internal meningeal layer of dura
- Separate regions of the brain



Dural Septa – reflections of the meningeal layer of dura that partially separate lobes of brain

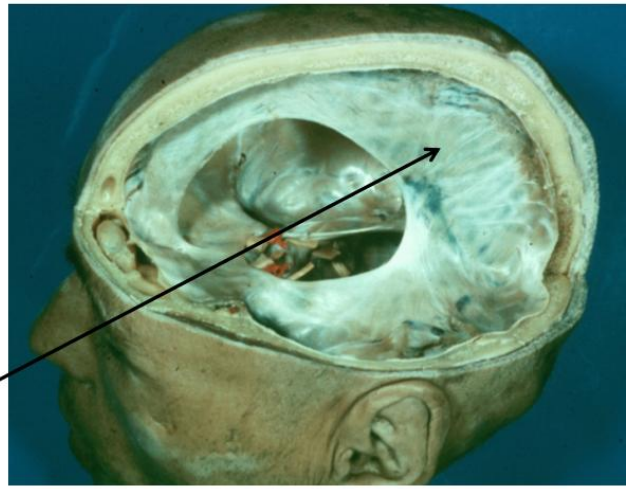
1. **Falx cerebri** (L. *falx* – sickle-shaped structure) – largest dural reflection, located in the median sagittal plane separating left & right cerebral hemispheres. Attaches to the frontal crest & crista galli anteriorly & the internal occipital protuberance posteriorly.
2. **Cerebellar tentorium** (Tentorium cerebelli; "tent of the cerebellum") – wide crescent- shaped dural fold separating the occipital lobes of the cerebrum from the cerebellum.
    a. Divides cranial cavity into ***supratentorial & infratentorial compartments***
    b. ***Tentorial notch*** – opening in anterior portion of cerebellar tentorium through which the brainstem (midbrain, pons, & medulla) passes

❖ **Clinical Notes:** In general, space-occupying lesions (tumors, arterial bleeding, blockage of CSF, etc.) in *children* occur ***infratentorially*** (below the cerebellar tentorium). In *adults*, most space-occupying lesions occur ***supratentorially*** (above the cerebellar tentorium). Supratentorial lesions can cause increased intracranial pressure & lead to herniation of the ***uncus*** (most medial part of the temporal lobe) inferiorly through the tentorial notch, called an **uncal herniation**. A patient with an **uncal herniation** has three cardinal symptoms: 1) a decreased level of consciousness (due to compression of the reticular formation in the midbrain), so the patient acts lethargic; 2) ipsilateral fixed dilation of the pupil, due to compression of the oculomotor nerve (CN III), which controls the pupillary sphincter as it exits anterior to the tentorial incisure from the midbrain; and 3) irregular breathing (initially hyperventilation, then hypoventilation with progressive herniation) and respiratory arrest secondary to compression of the respiratory center in the brainstem. If the intracranial pressure is not relieved, the patient can die due to **tonsillar herniation**, where the cerebellar tonsils push inferiorly into the foramen magnum, resulting in compression of the medulla & brain structures controlling vital functions (breathing, heart rate, etc.). (

LO#2

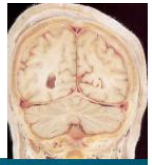
Cerebral falx – Dural Fold

- Separates right and left cerebral hemispheres
- Attaches to frontal crest, crista galli, internal occipital protuberance and midline of calvaria

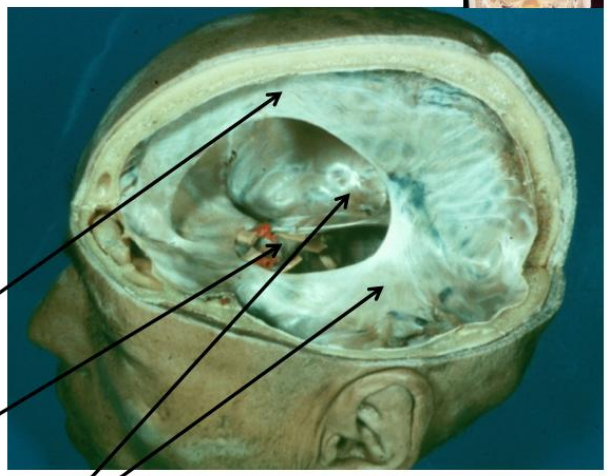


LO#2

Cerebellum tentorium – Dural Fold

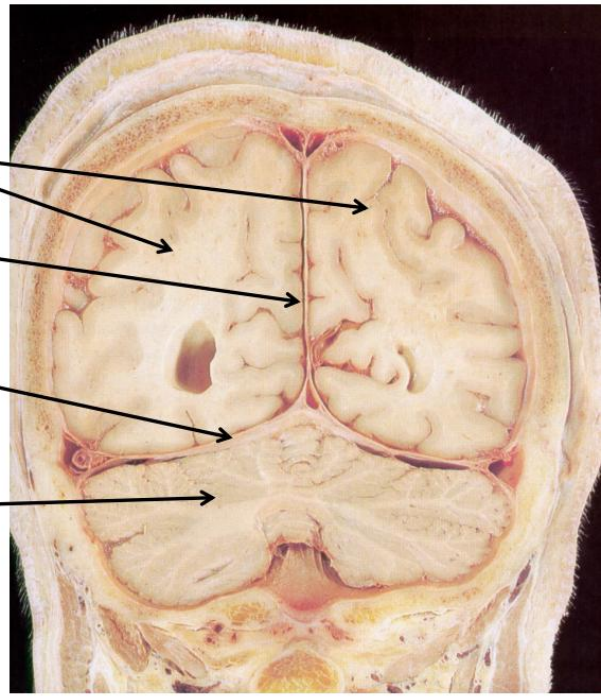


- Separates cerebellum from the cerebrum
- Continuous with cerebral falx
- Tentorial notch anteriorly allows passage of brainstem



LO#2

Dural Reflections

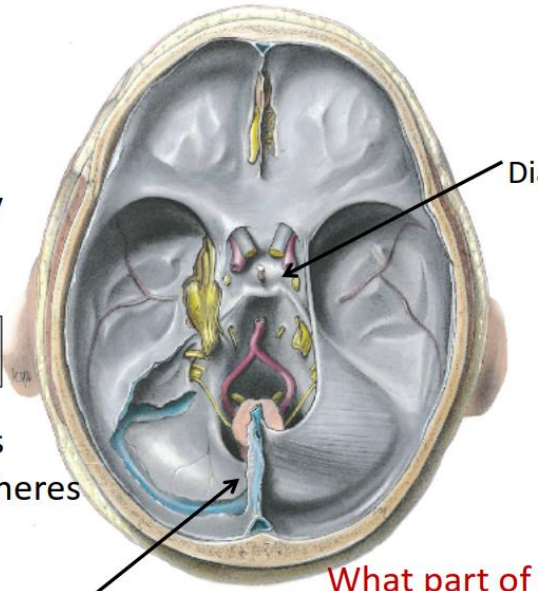


LO#2

Other Folds

Sella Diaphragm
- roof over pituitary
- small opening for infundibulum

Cerebellar Falx
- partially separates cerebellar hemispheres



What part of middle cranial cavity houses the pituitary gland?

c. **Falx cerebelli** (Cerebellar falx) – septa partially separating the cerebellar hemispheres. Lies inferior to the cerebellar tentorium in the posterior cranial fossa, where it attaches to the internal occipital crest. *Occipital sinus* is located in its base.

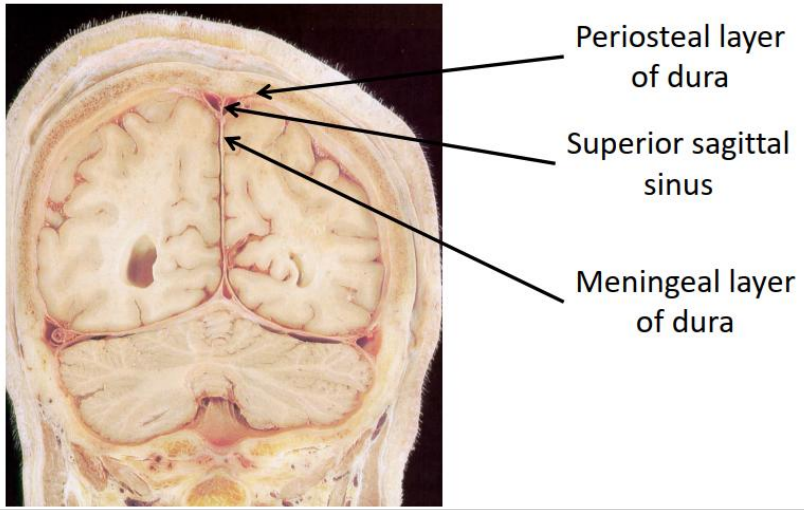
d. **Diaphragma sellae** (Sellar diaphragm) **– s**mall, circular sheet of dura suspended between the clinoid processes that covers the hypophyseal fossa in the sella turcica. The infundibulum of the pituitary gland penetrates this septum. (The infundibular stalk will break when the brain is removed in the lab. Therefore, the pituitary gland will remain inside the hypophyseal fossa of the sella turcica when the brain is removed.)

- **Clinical Note:** ***Pituitary tumors*** may extend superiorly through the aperture in the diaphragma sellae or cause this septum to bulge superiorly. Besides producing disturbances in endocrine function, these tumors often present with compression of the optic chiasm where the optic nerves cross, leading to *bitemporal hemianopsia* (or blindness in the temporal halves of the visual field).

LO#2

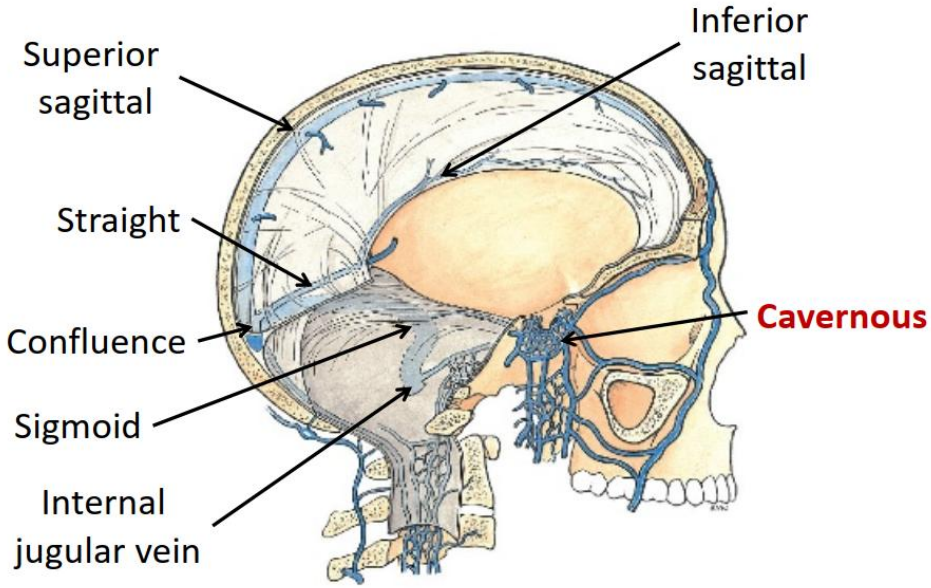
Venous Sinuses

- Located between periosteal and meningeal layers of dura
- Large veins from the brain empty into sinuses which drain into the internal jugular veins (IJV)



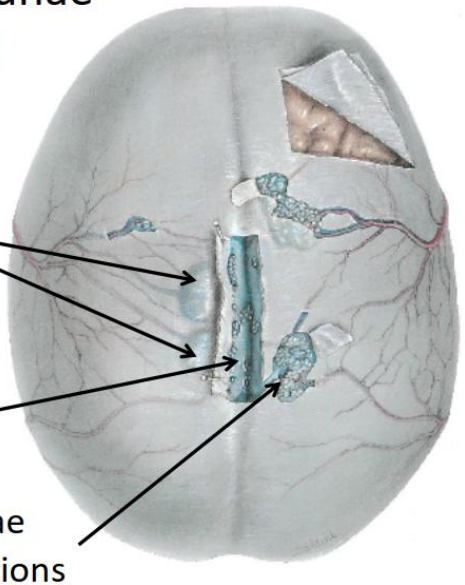
LO#2

Venous Dural Sinuses



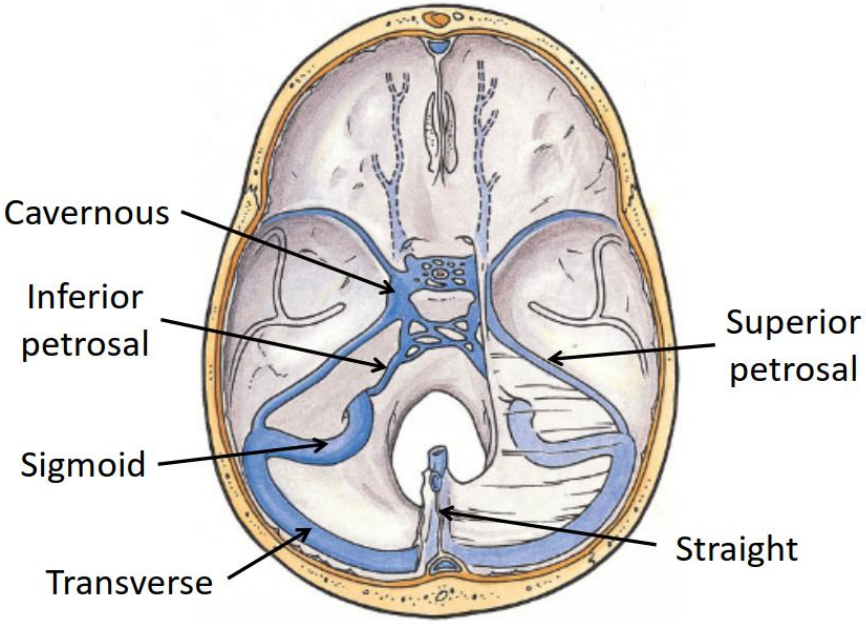
LO#2

Superior Sagittal Sinus, Lateral Venous Lacunae and Arachnoid Granulations



LO#2

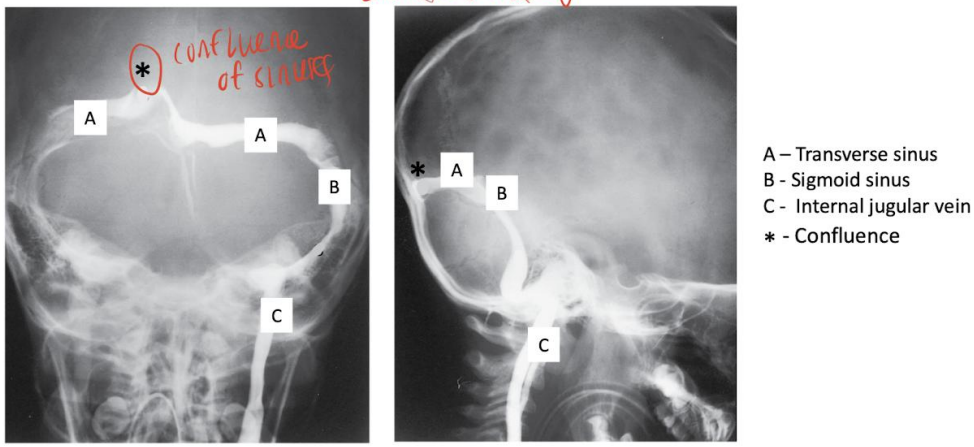
Venous Dural Sinuses



Venograph showing sinuses

contrast enhanced

Lateral view



(A) Anteroposterior view

(B) Lateral view

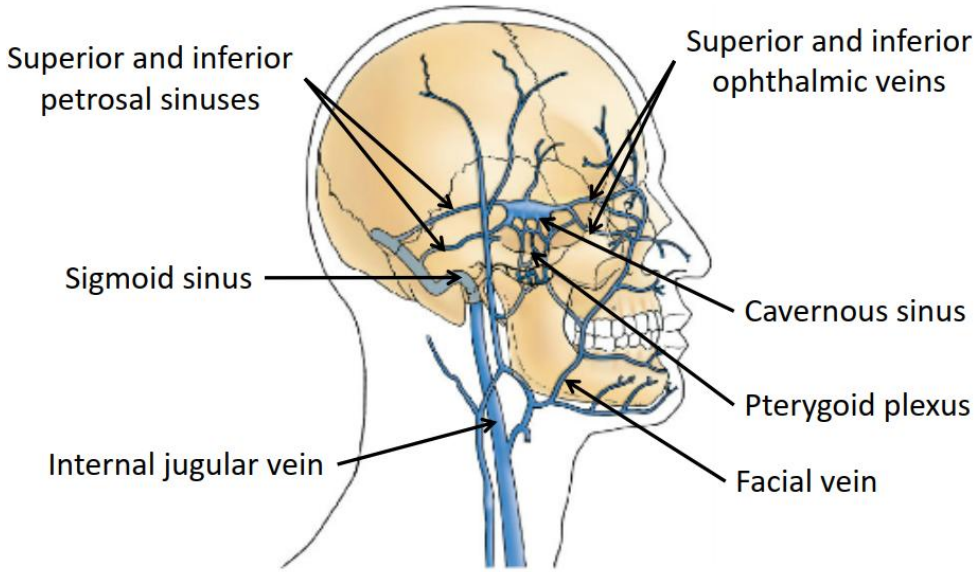
A – Transverse sinus
B - Sigmoid sinus
C - Internal jugular vein
∗ - Confluence

Dural venous sinuses – venous channels formed between two layers of dura

1. Large veins from the brain surface drain into these sinuses. Most of the blood from the brain drains through them into the *internal jugular veins*.
2. ***Arachnoid granulations*** (collections of arachnoid villi) are prolongations of arachnoid protruding through the meningeal layer of dura mater into the dural venous sinuses. They transfer CSF from the subarachnoid space to the venous system. They may erode bone, forming granular foveolae (pits) in the calvaria.
3. **Superior Sagittal Sinus –** lies in median plane in the superior border of the falx cerebri. It receives venous drainage from the ***superior cerebral veins***. It drains into the **confluence of sinuses**, a dilation of the venous channels located posteriorly, where the superior sagittal, straight, occipital & transverse sinuses meet.
    a. **Clinical Note:** The **superior cerebral veins** are clinically important as they can be torn following a blow to the head resulting in a **subdural hemorrhage** (bleeding into the potential space between the arachnoid & dura). (***COA p.876-7***)
4. **Inferior Sagittal Sinus –** occupies the inferior edge of the falx cerebri & ends in the straight sinus.
5. **Straight Sinus –** formed by the inferior sagittal sinus & ***great cerebral vein*** (of Galen). It runs along the line of attachment of the falx cerebri to the cerebellar tentorium to join the confluence of sinuses.
6. **Occipital Sinus –** lies within the attached border of the falx cerebelli. It drains superiorly into the confluence of sinuses. It also communicates inferiorly with the internal vertebral venous plexus.
7. **Confluence of Sinuses –** meeting place of the superior sagittal, straight, occipital, and transverse sinuses.
8. **Transverse Sinuses –** paired sinuses that pass laterally from the confluence of sinuses. They form deep grooves in the internal aspect of the skull. They become the sigmoid sinus at the posterior aspect of the petrous temporal bone as they leave the cerebellar tentorium.
9. **Sigmoid Sinuses –** paired sinuses that follow "S-shaped" courses in the posterior cranial fossa. They form deep grooves in the temporal and occipital bones. They turn anteriorly & continue inferiorly as the **internal jugular vein** upon crossing the ***jugular foramen***.

LO#2

Connections between venous sinuses and superficial veins of face



LO#2

Relations of Cavernous Sinus

Optic chiasm (II)

Internal Carotid Artery

Oculomotor nerve (III)

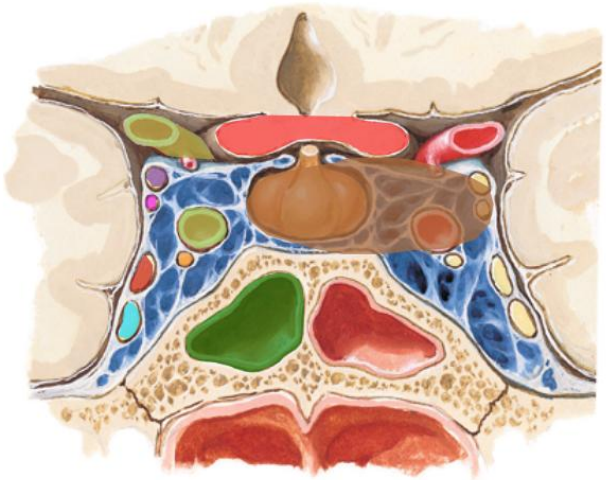
Trochlear nerve (IV)

Abducent Nerve (VI)

Ophthalmic nerve (V_1)

Maxillary nerve (V_2)

Sphenoidal sinus



Cavernous Sinuses – large, paired sinuses located on either side of the sella turcica, consisting of a plexus of thin-walled veins.

a. They receive blood from the superior & inferior ophthalmic veins, superficial middle cerebral vein, and sphenoparietal sinus. Cavernous sinuses communicate with each other through **intercavernous sinuses**. Blood drains out posteroinferiorly through the superior & inferior petrosal sinuses & emissary veins to the basilar and pterygoid plexuses.
b. **Clinical Note:** The facial & angular veins are connected by way of the ophthalmic veins to the cavernous sinus. Infection from the forehead, eyes, nose, & lips (*danger area of the face*) may spread to the cavernous sinus.
c. **CN III**, **CN IV**, **CN V1**, & **CN V2** are embedded along the *lateral wall* of the cavernous sinus. The **internal carotid artery** and **CN VI** run within the cavernous sinus.

Superior Petrosal Sinuses – paired sinuses that lie superior to the petrous ridge of the temporal bone in the attached margin of the tentorium cerebelli; they drain the cavernous sinuses and empty into the transverse sinuses.

Inferior Petrosal Sinuses – paired sinuses that drain the lateral cavernous sinus directly into the internal jugular vein on either side. The *basilar plexus* connects the inferior petrosal sinuses and communicates inferiorly with the internal vertebral venous plexus.

Emissary Veins – connect dural venous sinuses with veins outside the cranium (size & number vary)

a. **Clinical Note:** Infections within the scalp can spread to the bone via these veins or the foramina through which they pass. They usually carry blood *away* from the brain; however, they contain no valves. Therefore, blood can potentially flow in both directions.

Blood Supply to the Dura Mater

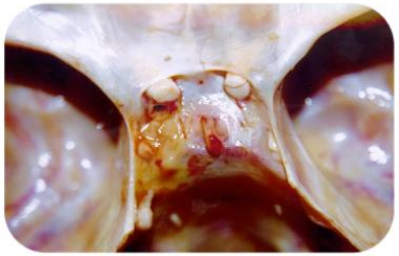
1. The blood supply to the dura is rather scant since the dura is relatively inactive metabolically; therefore, most of the blood in the "meningeal" arteries is distributed to the bones of the skull.
2. **Middle meningeal artery** – largest artery to the dura
    1. Branch off the maxillary artery (from the external carotid artery) and divides into *anterior & posterior branches*
    2. Enters the middle cranial fossa via the **foramen spinosum**, where it forms deep grooves in the inner lamina of bone, and is thus subject to injury.
    3. Supplies most of the supratentorial dura except the floor of the anterior cranial fossa (which is supplied by branches of the **anterior & posterior ethmoidal arteries** from the ophthalmic artery).
3. The infratentorial dura is supplied by branches of the **occipital artery** (a br. of the external carotid artery) as well as by the *anterior & posterior meningeal branches* of the **vertebral artery**.

LO#2

Cavernous Sinus Syndrome

Distinctive Presentation

- orbital or facial pain
- orbital swelling and chemosis
- fever
- oculomotor neuropathy
- trigeminal neuropathy



Most commonly results from cavernous thrombosis

- also carotid aneurysm, carotid-cavernous fistula, meningiomas, pituitary tumours

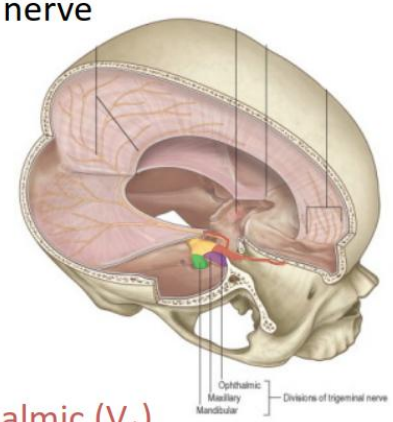


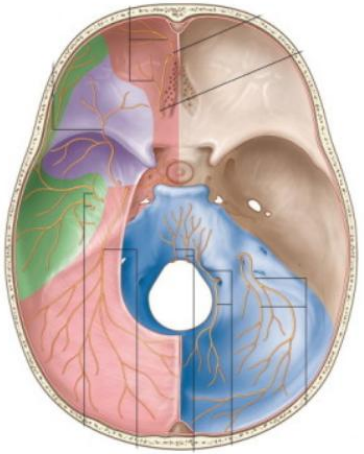


LO#3

Innervation of Dura Mater

Three divisions of trigeminal nerve





- Ophthalmic (V_1)
- Maxillary (V_2)
- Mandibular (V_3)



Innervation of the Dura Mater

a. In general, the dura is largely supplied by branches of the trigeminal nerve (CN V) supratentorially and the C2 & C3 nerves infratentorially (posterior cranial fossa).
b. However, the dura of the roof of the posterior cranial fossa (or tentorium cerebelli) is innervated by the tentorial branches of CN V1.

Clinical Note: The dura is sensitive to pain, especially where it is related to the dural venous sinuses & meningeal arteries. Consequently, pulling on arteries at the cranial base or veins near the vertex, where they pierce the dura, causes pain. Dural pain is usually referred & perceived as a headache arising in the cutaneous or mucosal regions supplied by the involved cervical nerve or division of the trigeminal nerve

Arachnoid mater – ("spider-like") intermediate layer closely held against the meningeal layer of dura by the pressure of the CSF.

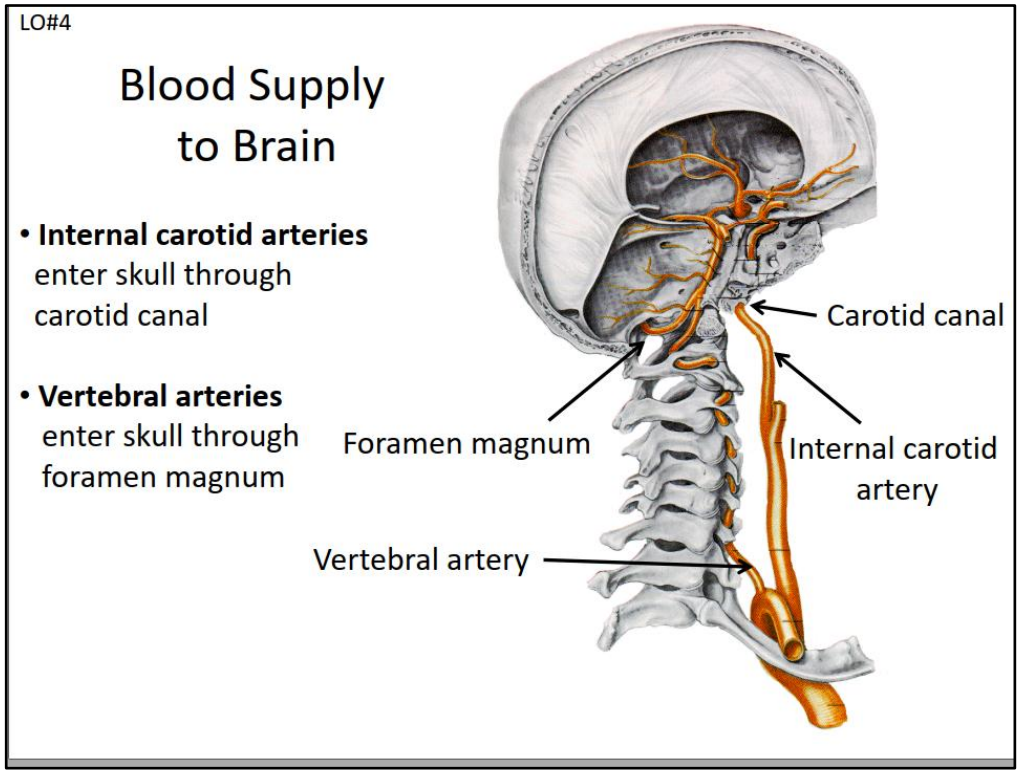
Pia mater – delicate, internal layer that adheres to the brain surface & follows all of its contours

Spaces between the cranial meninges

1. **Epidural space** – ***potential space*** between the periosteal layer of dura & calvaria. Can become an actual space if branches of the middle meningeal artery rupture (epidural hemorrhage), usually due to a blow to the head.
2. **Subdural space** – ***potential space*** between the meningeal layer of dura & arachnoid mater. Can become an actual space as a result of a blow to the head that causes dural venous blood to collect there (subdural hemorrhage).
3. **Subarachnoid space** – ***real space*** between arachnoid & pia mater, filled with ***CSF*** & contains ***blood vessels.*** Subarachnoid hemorrhages may occur in the absence of trauma, usually as the result of the rupture of an aneurism in an artery of the Circle of Willis.

Main Points – Venous Sinuses

- Dural reflections or folds are formed by folding together of inner meningeal layer of dura mater
- Dural folds separate regions of brain
- Pockets between 2 layers of dura mater created by folds form venous sinuses
- Venous sinuses drain cerebral veins through jugular foramen forming the internal jugular vein
- Venous sinuses also collect and drain excess CSF via arachnoid granulations
- Cavernous sinus is vulnerable to infection and is of clinical significance



VII. Arterial Supply to the Brain

A. Internal carotid arteries

1. Formed following the bifurcation of the common carotid artery (vertebral level C_4).
2. Enter the cranial cavity through the carotid canal in the petrous part of the temporal bones, along with the carotid plexus of sympathetic nerves.
3. Run toward & superior to the foramen lacerum, then travel anteriorly through the cavernous sinus toward the optic canal.
4. Give off the ophthalmic arteries, turn 180° posteriorly to travel over the cavernous sinus, and end as they provide anterior inflow to the ***Circle of Willis.***
5. Terminate as the ***anterior & middle cerebral arterie***s.

B. Vertebral arteries

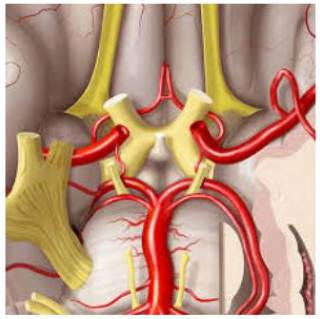
1. 1st branch off each subclavian artery.
2. Ascend through the transverse foramina of the first six cervical vertebrae. Run along the vertebral groove in the posterior arch of the atlas before piercing the atlanto-occipital membrane and entering the cranial cavity via the foramen magnum.
3. Run along the anterolateral surface of the medulla to join the other vertebral artery and form the basilar artery. The ***basilar artery*** gives off the ***posterior cerebral arteries*** on either side and a posterior communicating artery on either side to contribute posterior circulation to the Circle of Willis.

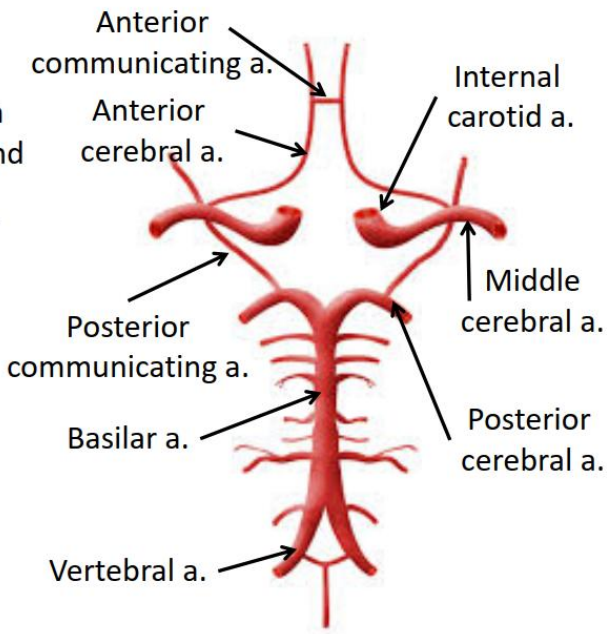
Clinical Note: **CN III** is located between the ***superior cerebellar*** and ***posterior cerebral arteries.*** Dilation (aneurysm) of either can compress and impair the function of CN III.

LO#5

Circle of Willis

- Important anastomoses on ventral aspect of brain
- Encircles pituitary stalk and optic chiasm
- Middle cerebral a. is NOT part of circle

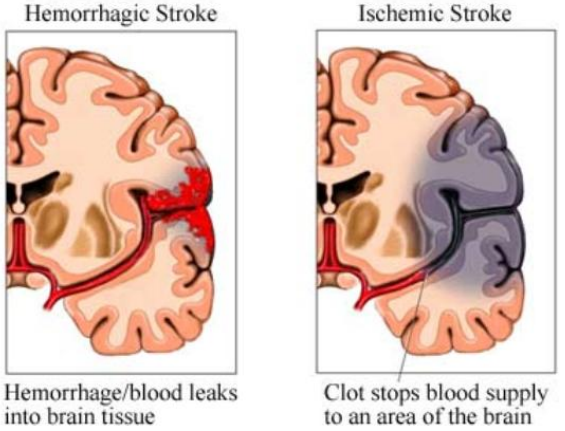




LO#5

Stroke

- Circle of Willis is efficient at making up for a gradual occlusion
- Sudden occlusion or a bursting of an aneurysm results in some neurological deficit



Nucleus Medical Media

Cerebral arterial circle (Circle of Willis)

a. Important anastomosis between the ***2 vertebral (1 basilar) & 2 internal carotid arteries*** at the base of the brain. The arterial vessels form a concentric ring around the sella turcica.
b. Formed from ***anterior to posterior*** by the:
   i. Anterior communicating artery
   ii. Anterior cerebral arteries
   iii. Internal carotid arteries
   iv. Posterior communicating arteries
   v. Posterior cerebral arteries

LO# 6 & 7



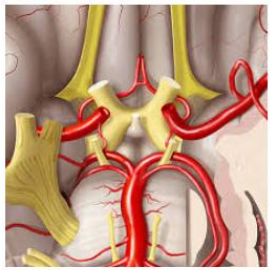
Anterior Middle Posterior

Anterior Cerebral Artery

- Enters longitudinal fissure and curves posteriorly around corpus callosum
- Supplies medial surfaces of frontal and parietal lobes, olfactory bulb/tract and corpus callosum and superolateral surface of frontal and parietal lobes

Middle Cerebral Artery

- Terminal branch of internal carotid artery
- Enters lateral fissure
- Supplies lateral surfaces of frontal, parietal and temporal lobes



Posterior Cerebral Artery

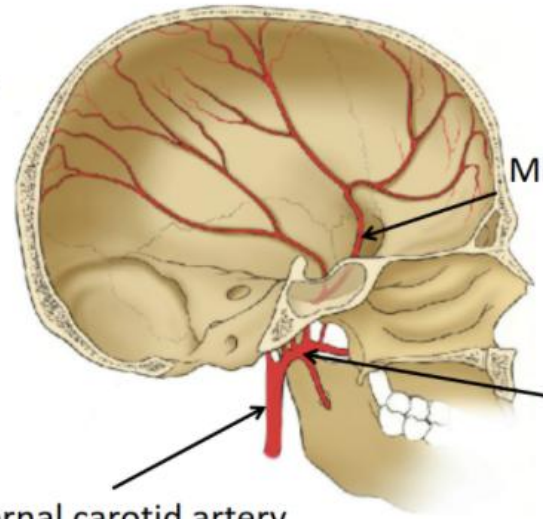
- Terminal branch of basilar artery
- Curves dorsolaterally around midbrain
- Supplies medial and inferior surfaces of temporal and occipital lobes

LO#8

Middle Meningeal Artery



Supplies meninges



Middle meningeal artery

Maxillary artery

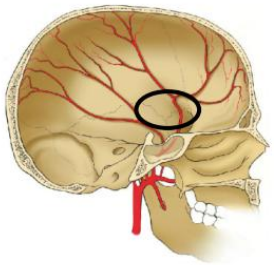
External carotid artery

Main Points – Blood to Brain

- Blood reaches brain via left and right vertebral arteries and left and right internal carotid arteries
- Vertebral arteries arise from subclavian arteries and ascend through transverse foramina in cervical vertebrae – enter skull through foramen magnum
- Internal carotid arteries arise from common carotid arteries and enter skull through carotid canals
- Branches of internal carotid and vertebral arteries form Circle of Willis
- Circle of Willis is found on ventral surface of brain around the pituitary stock and provides collateral routes for blood to cerebrum

LO#8

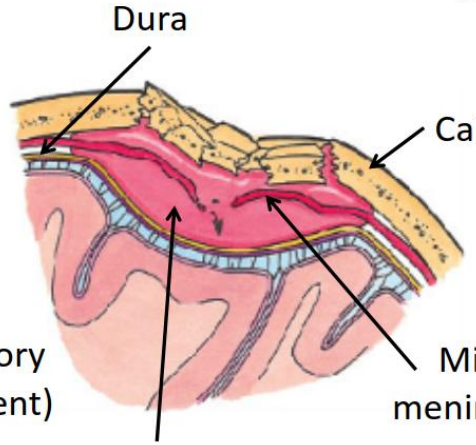
Epidural Hematoma



Fracture at pterion can tear middle meningeal a.

Clinical Course

- brief LOC followed by lucid interval
- compression of brain results in coma
- herniation of central structures and respiratory failure (without treatment)



Imaging of Epidural Hematoma

Lens-shaped



Normal



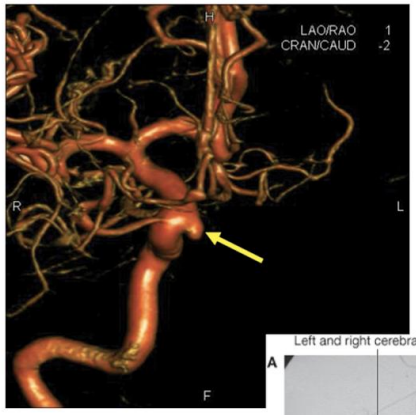
Epidural Hematoma

Epidural hemorrhage: Occurs when blood collects between the ***skull*** & the ***periosteal layer of dura.*** It often results from a hard blow to the side of the head near the ***pterion,*** which can fracture the squamous portion of the temporal bone and lacerate the ***middle meningeal artery*** (or veins).

- Since it is usually arterial in origin, it is under high pressure and blood accumulates rapidly.
- Typically, a ***brief concussion*** (loss of consciousness) occurs followed by a lucid interval of some hours. Later, drowsiness and coma occur.
- Compression of the brain occurs as the hematoma increases, necessitating evacuation of the blood and occlusion of the bleeding vessels.
- On radiographs of the head, this type of bleeding takes on a ***"lens-shaped"*** appearance because the blood is pressing against the skull and meninges

LO#2

Aneurysm of internal carotid

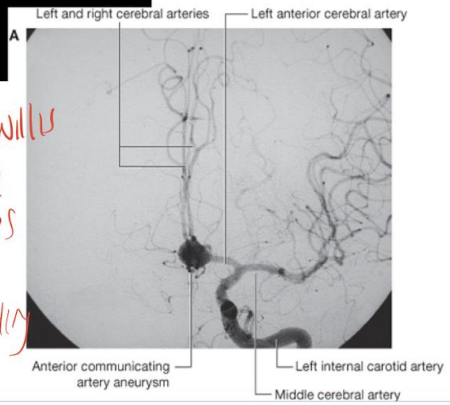


Weakening of vessel walls such that you have bulb appearing
↓burst, leading to aneurysm

- Dilation of the blood vessel
- Dilated area can burst

What might happen if internal carotid burst?

influences Circle of Willis
- anterior/middle cerebral arteries come off internal carotid → devastating



Subdural hemorrhage: Occurs between the ***dura mater*** and the ***arachnoid*** and is venous in origin (usually a tributary to one of the large dural veins or ***superior cerebral veins***).

- This type of hemorrhage is more common in ***older patients*** due to decreased brain volume and increased vein wall fragility with age.
- Bleeding in this type of injury is ***slow*** (compared to arterial bleeds) and may not come to clinical attention for weeks or months after the original injury.
- Symptoms are often mild and amorphous, but may include memory loss, altered mental state, and signs of increased intracranial pressure.
- On radiographs, this type of bleed appears ***crescent-shaped.***

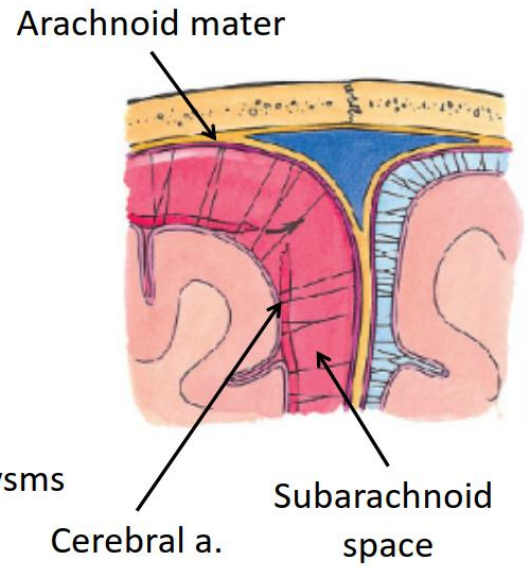
LO#2

Subarachnoid Hematoma

Rupture of cerebral a. in subarachnoid space

Symptoms
- "Thunderclap" headache
- nausea, vomiting & LOC
- neck stiffness
- visual changes

Clinical Considerations
- most commonly caused by saccular or berry aneurysms
- blood in CSF



Imaging of Subarachnoid Hematoma



Normal



Subarachnoid Hematoma

Subarachnoid Hemorrhage: Results from a blood vessel rupture within the ***subarachnoid space*** (where the CSF travels).
- It is often due to rupture of an aneurysm (dilation of an artery), such as an aneurysm of the internal carotid artery or Circle of Willis.
- Rupture of a vessel in the subarachnoid space will turn the CSF red or pink. A patient with a subarachnoid hemorrhage will present with the ***three cardinal symptoms***: a sudden onset of the "worst headache of his/her life", nuchal rigidity (stiff neck), and loss (or a decreased level) of consciousness.

Cerebrovascular Accidents: Anytime the blood supply to a portion of the brain is interrupted, it is known as a ***cerebrovascular accident (or CVA)***, or ***stroke***.
- There are two main types of stroke, **hemorrhagic** & **ischemic**.
  - In a ***hemorrhagic stroke***, a vessel within the parenchyma of the brain ruptures. This prevents blood from going to its normal location and results in decreased perfusion for some portion of the brain.
  - In an ***ischemic stroke,*** the blood supply to a portion of the brain becomes blocked preventing blood from reaching its destination.
  - Ischemic events may be ***thrombotic*** (a blood clot forming at the location) or ***embolic*** (a clot forming at a distant location travels to the brain).

The location of the blockage or bleed determines the symptoms, which are typically *sudden* in onset and affect regions of the body controlled by the affected portion of the brain. Symptoms may include weakness, paralysis, sensory deficit, language abnormality, speech abnormality, or memory loss

Main Points – Cranial Hematomas

Epidural hematoma

- Blood in epidural space (between dura mater and skull)
- Caused by ruptured middle meningeal artery often at pterion

Subarachnoid hematoma

- Blood in subarachnoid space (between arachnoid mater and pia mater)
- Caused by ruptured cerebral artery
- Often due to aneurysm

Subdural hematoma

- Blood (venous) in subdural space (between dura mater and arachnoid mater)
- Caused by tear in venous sinus
- Due to rapid acceleration or deceleration of brain in skull

IX. Overview of the Cranial Nerves

A. **Cranial nerves** (**CN**) are bundles of nerve fibers that **exit** the cranial cavity (**12 pairs**). They are identified by name &/or a Roman numeral (CN I – XII) from rostral to caudal based on their origin off the brain. Some cranial nerves carry purely *sensory fibers*, others are purely *motor*, & several are *mixed*. Cranial nerve exams are part of routine physical & neurological exams.

B. Intracranial Path of the Cranial Nerves

1. ***Olfactory nerves (CN I)*** *– special sense of smell*
    1. *Axons from the olfactory areas of the nasal cavities traverse **cribriform foramina** of the **cribriform plate** (ethmoid bone) to synapse on neurons in the **olfactory bulbs** of the brain.*
    2. *Axons of these neurons form **olfactory tracts** that terminate at the cerebral cortex.*
2. ***Optic nerve (CN II) –*** *special sense of vision*
    1. *Extends posteriorly from eyeball to exit through the **optic canal** & enter the middle cranial fossae.*
2. *Fibers decussate (cross) in the **optic chiasm** & join uncrossed fibers to form the **optic tract**. Fibers in the optic tract terminate in the lateral geniculate bodies of the thalamus.*
3. ***Oculomotor nerve (CN III) –*** *motor*
    1. *Originates from midbrain, pierces the dura lateral to the diaphragma sellae, to run through the lateral wall of the cavernous sinus.*
    2. *Enters the orbit via the **superior orbital fissure**, dividing into muscular branches to levator palpebrae superioris, and 4 extraocular muscles (superior rectus, medial rectus, inferior rectus, & inferior oblique).*
3. *Carries parasympathetic fibers to the sphincter pupillae muscle (constricts pupil) and the ciliary muscle/body (accommodation).*
4. ***Trochlear nerve (CN IV) –*** *motor*
    1. *Runs the longest intracranial course as it originates from the **posterior** side of midbrain. Pierces dura near posterior clinoid process & runs along lateral wall of cavernous sinus.*
    1. *Enters the orbit through the **superior orbital fissure** to supply superior oblique muscle.*
5. ***Trigeminal nerve (CN V) –*** *motor & sensory*
    1. *Emerges from brainstem by a large sensory root & small motor root. The sensory root forms the **trigeminal ganglion** (where its cell bodies are located) under the dura. The motor root of CN V runs parallel to the sensory root, bypasses the ganglion, & becomes part of the mandibular nerve (CN V_3).*
    2. *Trigeminal ganglion forms 3 nerves/divisions:*
        1. ***Ophthalmic nerve (CN V_1) –*** *Exits cranial cavity via **superior orbital fissure**. **Sensory** distribution to the skin of forehead, scalp, eyelids, nose; cornea; paranasal sinus & superior nasal mucosa.*
        2. ***Maxillary nerve (CN V_2) –*** *Exits cranial cavity via **foramen rotundum**. **Sensory** distribution to the skin of the upper jaw, including the upper lip, maxillary teeth, inferior nasal mucosa, maxillary sinus, and palate.*
        3. ***Mandibular nerve (CN V_3) –*** *Exits cranial cavity via **foramen ovale**. **Sensory** distribution to the skin over the mandible, including the lower lip, buccal, parotid, & temporal regions of face, external ear, mandibular teeth, temporomandibular joint, mucosa of mouth, and anterior 2/3 of tongue (touch & temperature). **Motor** to muscles of mastication, mylohyoid, ant. belly of the digastric, & tensor veli palatini.*
6. ***Abducent nerve (CN VI)*** *– motor*
    1. *Emerges from the brainstem between the pons & medulla. Runs longest intradural*

course of the cranial nerves, piercing the dura on the clivus, then coursing through the middle of the cavernous sinus.

2. *Exits cranial cavity through the **superior orbital fissure** to enter the orbit to supply the lateral rectus muscle.*

7. ***Facial nerve (CN VII) –** motor & sensory*
    1. *Arises from brainstem and exits cranial cavity via the **internal acoustic meatus**. It runs through the **facial canal** in the petrous part of the temporal bone and exits the cranium via the **stylomastoid foramen** (longest intraosseous course).*
    2. *Carries motor (muscles of facial expression, stylohyoid, posterior belly of digastric), sensory, special sensory (taste from anterior 2/3 of tongue), and parasympathetic fibers.*
8. ***Vestibulocochlear nerve (CN VIII) –** special senses of hearing, equilibrium, & motion*

a. Originates from the brainstem and enters the **internal acoustic meatus**, where it divides into the vestibular (equilibrium & motion) and cochlear nerves (hearing). CN VIII does
not exit the cranium.

9. **Glossopharyngeal nerve (CN IX) –** *motor & sensory*

1. Emerges from lateral aspect of medulla and exits cranial cavity though **jugular foramen**.
2. Carries motor (*stylopharyngeus muscle*), *sensory* (from pharynx, carotid sinus & body), special sensory (*taste* from *posterior 1/3 of tongue*), and parasympathetic fibers

10. **Vagus nerve (CN X) –** *motor & sensory*

1. Arises from the brainstem and leaves the cranial cavity through the **jugular foramen**.
2. Runs *longest course* with the most extensive distribution of all the cranial nerves.
3. Carries *motor* (pharyngeal, palatine, most intrinsic larynx muscles), *sensory* (inferior pharynx & larynx), special sensory (*taste* from *root of tongue*), & *parasympathetic* fibers.

11. **Spinal accessory nerve (CN XI) –** *motor*

1. Arises from the ***superior 5-6 cervical segments of the spinal cord***, ascends to enter the cranial cavity via the **foramen magnum**, and then exits via the **jugular foramen**.
2. *Motor to trapezius and sternocleidomastoid.*

12. **Hypoglossal nerve (CN XII) –** *motor*

1. Arises from medulla and leaves the cranial cavity through the **hypoglossal canal**.
2. Motor to intrinsic & extrinsic tongue muscles (*styloglossus*, *hyoglossus*, & *genioglossus*)
3. Cervical spinal nerves “hitch a ride” with CN XII to the superior root of the ansa cervicalis to get to the infrahyoid muscles.

Name of Foramina	Location	Contents
Anterior Cranial Fossa		
Foramen cecum	Btwn ethmoid & frontal bones	Nasal emissary vein
Cribriform foramina	Ethmoid bone	Olfactory nerves (CN I)
Anterior & posterior ethmoidal foramina	Ethmoid bone	Anterior & posterior ethmoidal vessels & nerves
Middle Cranial Fossa		
Optic canal	Sphenoid bone	Optic nerve (CN II) Ophthalmic artery Central artery & vein of retina
Superior orbital fissure	Sphenoid bone	CN III, IV, V_1, VI Ophthalmic veins
Foramen rotundum	Sphenoid bone	Maxillary nerve (CN V_2)
Foramen ovale	Sphenoid bone	Mandibular nerve (CN V_3) Accessory meningeal artery
Foramen spinosum	Sphenoid bone	Middle meningeal vessels Meningeal branch of CN V_3
Foramen lacerum	Btwn sphenoid & temporal bones ***Closed by cartilage in life***	*Internal carotid artery** (*passes over)* Deep petrosal nerve Sympathetic fibers
Posterior Cranial Fossa		
Jugular foramen	Between temporal & occipital bones	CN IX, X, XI Internal jugular vein
Hypoglossal canal	Occipital bone	Hypoglossal nerve (CN XII)
		Spinal accessory nerve (CN XI)